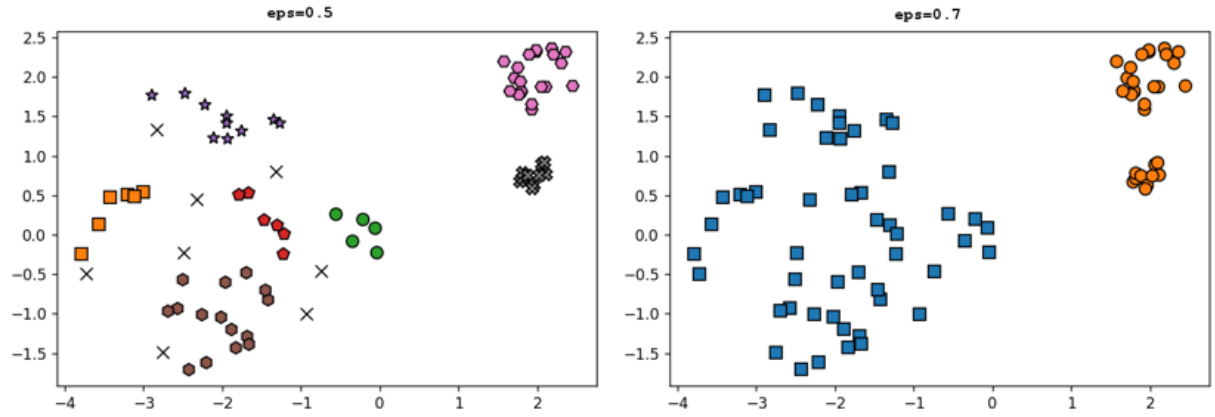


Figure 12.14 An example of a dataset with areas of heterogeneous density for which it's not possible to find a single value of ε that fits all areas. The dataset has a low-density cluster in the left area and two high-density clusters, close to each other, in the top-right corner.
(left) With a low value of ε, the low-density cluster is broken into many small clusters surrounded by noise. (right) With higher values, the two clusters on the right are merged. There is no value for which these clusters are correctly separated and at the same time the one on the left is recognized as a single cluster.

In the next couple of sections, we will examine two different ways to cope with hyperparameters and address their issues.

12.4 OPTICS

As we have seen in the last section, DBSCAN is a powerful algorithm able to identify non-linearly separable clusters of any shape; it has, however, a weak spot connected to the parameters regulating the density thresholds: in particular, it